

9/24 ch. 3.1

Bit: (Binary digit) The smallest unit of data that can be store in a computer and have two value of 0 or 1.

Byte: 8 bits equal to 1 byte.

Bit pattern: A sequence and often called a string of bits. Use it to represent different types of data. A bit pattern is made of 16 bits.

* With different processor represent different data type, memory stores them without recognizing datatype.

Data compression: To occupy less memory space, data is normally compressed before being stored.

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Sampling (取樣): Select only a finite number of point on the analog signals measure and record the value.

Quantization (量化): A process that round the recorded value of sample to the closet integer value.

Encoding: Quantized sample values need to be encoded as bit pattern

Bit depth: Bit per sample.

Bit rate (bit per sec): $\text{Bit rate} = \text{sampling rate} \times \text{bit depth}$

64 000 per second = 40000 samples per second and 16 bits per sample.

ch.3.5

Raster graphic

Resolution: Scanning rate in image processing. We need to decide how many pixels.

Color depth = Number of bits used to represent a pixel.

True - color

Index - color

Vector graphic

ch.3.6

frame: (幀)

ch.4.1